

GAYATHRI BALU

Junior Data Analyst,Aurora,IL

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EDUCATION

Anna University,Chennai

Bachelor of Engineering in Computer Science

Sep. 2008– May 2012

India

RELEVANT COURSEWORK

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|---------------------------------------|-----------------------------|-----------------------------------|-----------------------------|
| • Data Analysis & Visualization | • Statistics & Probability | • Exploratory Data Analysis (EDA) | • Pipelines |
| • Python Programming for Data Science | • SQL & Database Management | • Data Cleaning & ETL | • Healthcare Data Analytics |

EXPERIENCE

NumPy Ninja – Healthcare Analytics

Aug 2024

Junior Data Analyst (Volunteer)

Dover, DE

- Developed automated ETL workflows to extract, clean, and transform healthcare datasets from APIs and flat files, improving data readiness for analysis.
- Incorporated Python (Pandas, NumPy) and SQL scripts to validate data quality, aggregate records, and load curated datasets into PostgreSQL.
- Utilized Tableau and Power BI to build interactive dashboards that visualize KPIs, trends, and clinical metrics for stakeholder review.
- Automated recurring data preparation and reporting tasks, reducing manual effort and turnaround time.
- Collaborated in Agile/Scrum teams to deliver data-driven insights during datathons and sprint reviews.

PROJECTS

Sepsis Data Analysis Project | Python, Pandas, Tableau, NumPy

May 2025 – July 2025

- * Analyzed 1.5M+ records across 40,336 patient cases to track sepsis progression using biomarker and organ-level features.
- * Performed data cleaning, validation, and correlation analysis with Python (Pandas) to ensure accurate clinical insights.
- * Focused on SOFA biomarkers and lung function metrics to identify early indicators of sepsis onset.
- * Conducted comparative analysis between MICU and SICU cases to identify patterns and risk differences.
- * Designed interactive Tableau dashboards to visualize trends in biomarkers, organ impact, and clinical indicators (Septic Shock, MOD, ABG, APACHE, SIRS).
- * Collaborated in six Agile sprints to integrate data exploration, organ analysis, and dashboard development into actionable clinical insights.

Early Detection of Prediabetes Using CGM & Wearables | Python, SQL, Pandas, Multisensor Analysis

Nov 2025

- * Integrated Dexcom CGM and wearable wristband data (HR, IBI, EDA) over six months for early prediabetes detection.
- * Performed data cleaning, normalization, and outlier handling to prepare datasets for analysis.
- * Calculated glucose spike deltas and trends using Python and SQL to identify potential risk indicators.
- * Visualized glucose variability and lifestyle correlations in dashboards to support actionable insights for health monitoring.

Maternal Health & Abdominal Fat Assessment Study | Python, Power BI, ICD, SNOMED, RxNorm

Sep 2025

- * Managed and cleaned a 116-variable clinical dataset, including demographics, ultrasound measurements, nutrition, labs, and newborn outcomes.
- * Applied ICD and SNOMED codes to categorize maternal metabolic conditions and obstetric diagnoses.
- * Analyzed LOINC-coded lab results (glucose, hemoglobin A1c) to assess maternal and neonatal risk factors.
- * Evaluated RxNorm-coded medications for maternal substance use and metabolic management.
- * Developed interactive Power BI dashboards to visualize trends and support targeted prenatal care interventions.

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL(MySQL,PostgreSQL)

Data Analytics & BI Tools: Tableau, Power BI, Excel, SSRS, Streamlit

Databases & ETL: PostgreSQL, SSMS, SSIS, ETL Pipelines

Healthcare Data Standards: FHIR, OMOP, ICD-10, SNOMED CT, LOINC, RxNorm, HIPAA-compliant data handling

Methodologies & Processes: Agile/Scrum, JIRA, Predictive Analytics, Statistical Modeling, Healthcare Operations Analysis

Developer Tools & Platforms: GitHub, Git Bash